Arjun | Doctor Knowledge & Productivity Assistant

Health-a-thon 2026

About the Team / Organisation

A. Team / organisation: Arjun. Viren Mehta (Team Lead), Karan Patel and Shrey Patel from IIT Bombay; Perin Modi from NIT Calicut

B. Doctor background and role in the solution: Dr Akash Datt, MBBS (Sapthagiri Institute of Medical Sciences); MD in Radiation Oncology (Ramaiah University of Applied Sciences); now at IQVIA. **Medical partner:** gives the team practical and theoretical insight into the medical domain and where scalable solutions fit, and keeps the design secure, accurate and safe.

Problem Statement

Oncology knowledge is vast, changes fast and sits in English textbooks and PDFs. **Oncologists need a checkable answer in the middle of a clinic; patients and families need the same facts in their own language.**

1.56 M
new cancer cases in India in 2024, about 0.9 M deaths [1]

~1 : 1,500
oncologists to patients in India; about 1 : 100 in the US [2]

~10 %
of Indians report speaking English; 22 scheduled languages [3]

Doctors lose time. Answers are spread over several textbooks and guidelines, and typing long case notes is slow.

Patients are left out. The best patient guides (NCCN) are English PDFs; many families cannot read them.

Chatbots cannot be trusted blindly. They sound sure but give no page-level source and can invent facts, which is unsafe in cancer care.

[1] IARC GLOBOCAN 2024 (via WHO 2026, ORF) [2] EY 2015 via IndiaSpend, estimates vary [3] Census of India 2011

How a question becomes a cited answer (purple: online, teal: on our machine)



AI Solution Description

A voice-first, source-grounded oncology assistant. Doctors and patients speak or type in their own language. The AI searches indexed textbooks and guidelines, writes an answer only from what it finds, and cites the book and page, or says **"not available"**.

Faster for doctors: one spoken question replaces a search across four books; a cited answer in about 34 s on a CPU.

Safer for patients: plain-language answers in their language from NCCN guides, with emergency and crisis screening.

Technology and Tools Used

- **Qwen 2.5 / 3.5** local LLMs
- **llama.cpp** model server
- **BGE-base** embeddings
- **FAISS** vector index
- **BM25** keyword index
- **MiniLM** reranker
- **Streamlit** web app
- **Python** backend
- **Sarvam AI** voice, translate
- **scite.ai** literature
- **Tailscale** public link
- **SQLite** cache

Key Features of the solution

- **Voice-first for doctors.** Speak a clinical question in English or Hindi/Hinglish; what it heard is shown back; same cited answer as typing.
- **Cited textbook answers.** 4 textbooks, 19,949 passages; each fact opens the source page; it refuses when the books do not cover it.
- **Treatment Guidance.** A case in one sentence returns ranked ASCO recommendations (19 guidelines) with strength and evidence.
- **Patient Portal, voice in 11 languages.** NCCN-based answers in the patient's language and English, read aloud.
- **Research + Content Studio.** scite.ai literature answers; draft handouts and articles.

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Build Sprint Progress

Four weeks, from a working core to a public, voice-first product

WEEK 1 Foundation

- Textbooks extracted, chunked and indexed (BM25 + vectors + reranker)
- Local Qwen served on CPU; cited answers with page viewer
- Refusals when sources do not cover a question

Refined: *Chose the 3B model: the 1.5B one copied sources instead of answering.*

WEEK 2 Scale and productivity

- Full-book index: 19,949 passages from 4 textbooks
- Content Studio and scite.ai Research Platform
- Oncologist's 77 case questions run end to end

Refined: *Indexing whole books ended wrong refusals of in-scope questions (0/12).*

WEEK 3 Clinician + patient tools

- Treatment Guidance: 19 ASCO guidelines, 443 recommendations
- Patient Portal: 76 NCCN guides, Learn mode, safety screen
- Textbook fallback when no guideline matches

Refined: *Tested on unseen cases (5/10) and fixed the gaps found.*

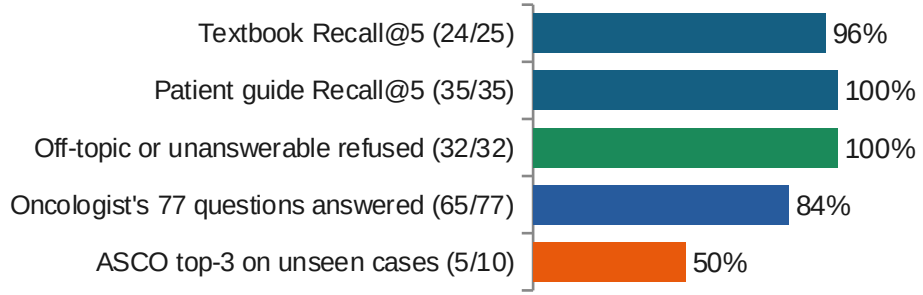
WEEK 4 Voice-first and launch

- Voice in 11 languages (Sarvam), 30 s cap, recording indicator
- Light/dark, phone-ready UI; public HTTPS link
- 124 pipeline and 95 UI checks; spend cap and key rotation

Refined: *Cached translations and audio: repeat use costs nothing.*

User Validation and AI Evaluation

77 clinical questions from our oncologist partner were run end to end through the solution.



Refusals worked: textbooks 10/10 off-topic, 8/8 unanswerable; patient portal 8/8 and 6/6; 12/12 safety cases routed.

Speed: about 34 s per answer on a CPU-only machine.

Limits: 12 of 77 declined (some long case vignettes); ASCO matching is rules-based.

Safety and Responsible AI

Answers only from sources. Three guardrails (topic gate, evidence threshold, citation check) refuse rather than guess; ASCO text is quoted verbatim (1,264/1,264 matched).

Anonymised before it leaves. A backend guardrail anonymises patient details (name, age, address, dates, IDs) in Research questions and answers before anything reaches scite.ai. Everything else runs on our own machine.

Patient safety screen. Crisis wording gets helpline numbers instead of an answer; emergency symptoms show an urgent-care notice; treatment decisions go back to the care team.

Human in the loop. Decision support, not medical advice.

Not yet: the same scrub for voice and translation text sent to Sarvam; bare names are not caught.

Deployment and Sustainability Plan

Live on the internet over HTTPS; anyone can use it, no install or login.

Credible sources: DeVita, Gunderson & Tepper, MD Anderson handbook, Oncologic Imaging, 19 ASCO guidelines, 76 NCCN patient guides.

Low cost: open models on one CPU machine; voice spend is cached and capped.

Doctors · Healthcare staff · Patients

Works on laptop and mobile

Supporting Links

MVP: v.taila0f4fa.ts.net
Video demo walkthrough: Watch video
Other: github.com/viren711/healthathon